

AI/ML Task 2

Model Optimization & Performance Comparison

Feature Engineering, Multi-Model Comparison, and Model Selection on the California Housing Dataset

1. Methodology

- Building on the Task 1 Linear Regression baseline ($R^2 = 0.576$), this task applies a more rigorous workflow.
- The California Housing dataset (20,640 rows, 8 input features, target = median house value) was split into training (80%) and test (20%) sets before any preprocessing, to prevent test-set information from leaking into feature scaling.
- A StandardScaler was fit on the training features only, then applied to both sets.
- Three regression models were then trained and evaluated on identical data: Linear Regression (baseline), Ridge Regression (alpha = 1.0, L2-regularized), and a Decision Tree Regressor (max_depth = 5).
- Each model was scored on the held-out test set using RMSE and R^2 , and Train R^2 was also recorded to check for overfitting.

2. Results

Model	Train R^2	Test R^2	RMSE
Decision Tree	0.6377	0.5997	0.7242
Ridge Regression	0.6126	0.5758	0.7456
Linear Regression	0.6126	0.5758	0.7456

Key takeaways:

- The Decision Tree Regressor achieved the lowest RMSE (0.724) and highest test R^2 (0.600), making it the best-performing model.
- Linear Regression and Ridge Regression produced nearly identical results to each other and to the Task 1 baseline. This is expected, since StandardScaler does not change Linear Regression's predictions — it only rescales the coefficients.
- A light Ridge penalty (alpha = 1.0) had negligible effect here, because the dataset already has a strong linear signal from median income, leaving little for regularization to correct.

2.1 Visual Performance Validation

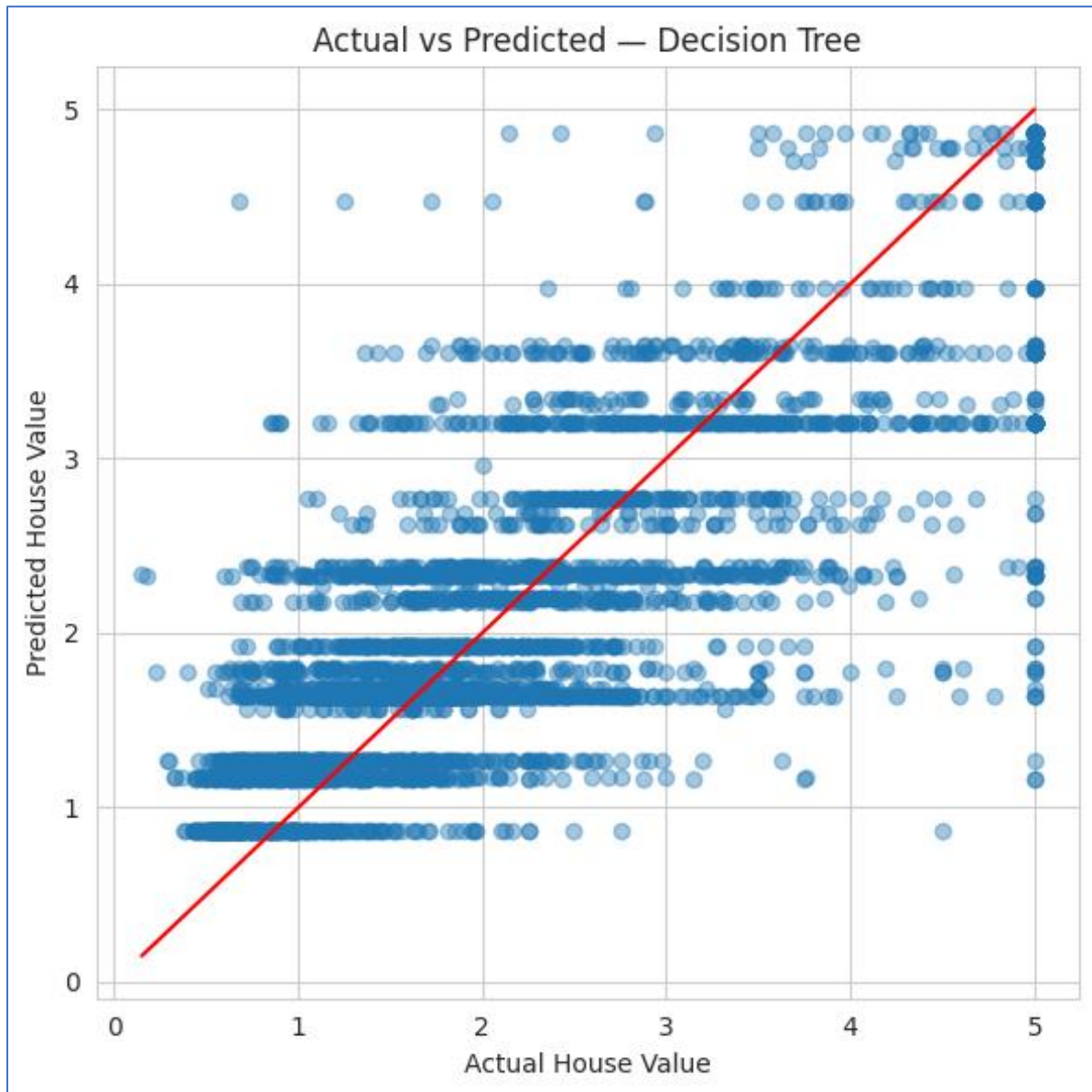


Figure 1 : Actual vs Predicted House Values — Decision Tree Regressor. Points closer to the red line mean better predictions.

- Predictions cluster tighter around the diagonal than in Task 1's Linear Regression plot
- A vertical band near the top is visible — this is the known \$500,000 price-cap artifact in the dataset, not a model flaw

3. Model Selection & Overfitting Check

Selected Model: Decision Tree Regressor (max_depth = 5)

The Decision Tree Regressor was selected as the final model based on its performance on the held-out test set, identified programmatically as the model with the lowest RMSE rather than chosen in advance.

Overfitting Analysis:

Model	Train R^2	Test R^2	Gap
Decision Tree	0.638	0.600	0.038
Ridge Regression	0.613	0.576	0.037
Linear Regression	0.613	0.576	0.037

All three models exhibit a small and comparable gap between training and testing performance, indicating that none of the models are overfitting to the training data. The Decision Tree's constrained depth (`max_depth = 5`) limits its complexity sufficiently to generalize well, while still allowing it to capture non-linear relationships — such as income thresholds and geographic clustering effects — that linear models are structurally unable to represent.

4. Conclusions

- The Decision Tree Regressor outperformed both Linear Regression and Ridge Regression, achieving an approximately 4% improvement in test R^2 (0.600 versus 0.576). This result supports the hypothesis that median house prices in the California Housing dataset are influenced by non-linear relationships among features, which linear models are inherently limited in capturing.
- Feature scaling did not materially alter the performance of the linear models in this instance, since Linear Regression is scale-invariant by construction and the applied Ridge penalty ($\alpha = 1.0$) was insufficient to meaningfully affect a dataset already dominated by a strong linear predictor (median income). Nonetheless, scaling remains a standard and necessary preprocessing step for regularized and distance-based algorithms more broadly.
- Overall, this task demonstrates a complete and methodologically sound machine learning workflow: proper preprocessing, comparative evaluation across multiple algorithms, and objective, metric-driven model selection — consistent with practices used in professional machine learning development.